C.E.L.L.: A Human-in-the-Loop Web Platform Integrating Blood Cell Morphology Education and Student-Generated Label Data

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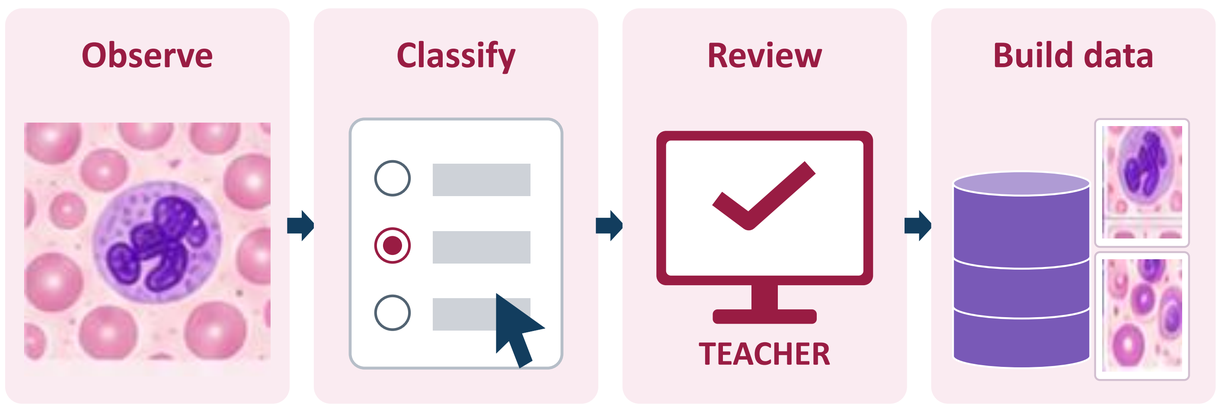
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01 BACKGROUND & OBJECTIVE

Learning responses as candidate labels

Blood cell morphology training requires repeated observation and feedback fro m teachers with expertise in blood cell morphology. Student classifications can also provide candidate labels for teacher review.

We developed C.E.L.L., a web platform that connects morphology education with the accumulation of verified label data.



02 C.E.L.L. PLATFORM

Independent classification

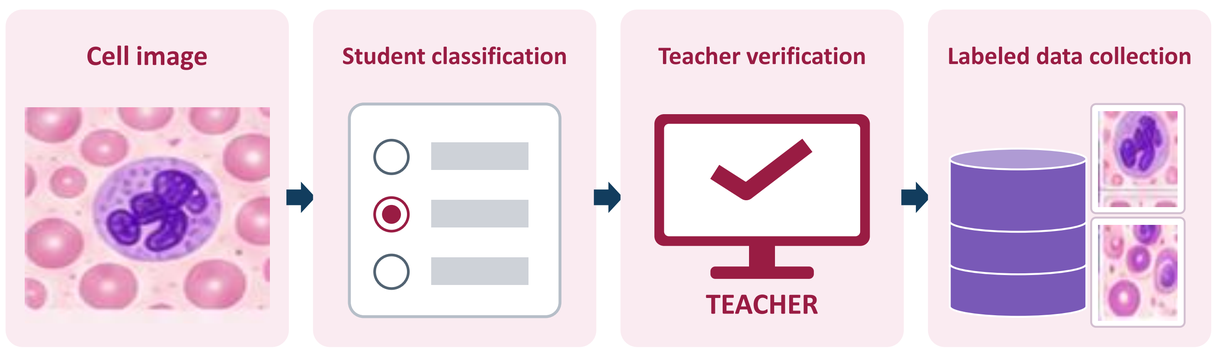
Students classify cells before viewing reference labels or AI predictions.

Learning feedback

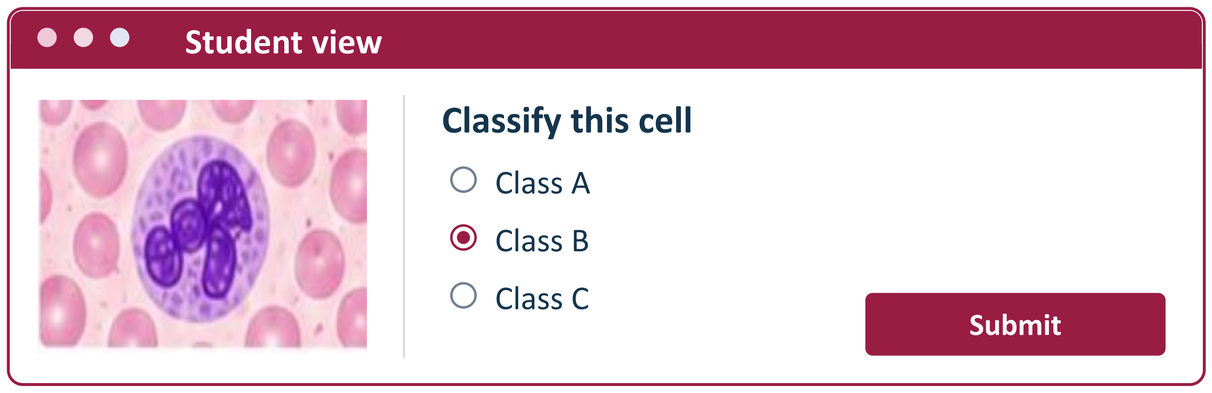
Progress, accuracy and confusion matrices support learner assessment.

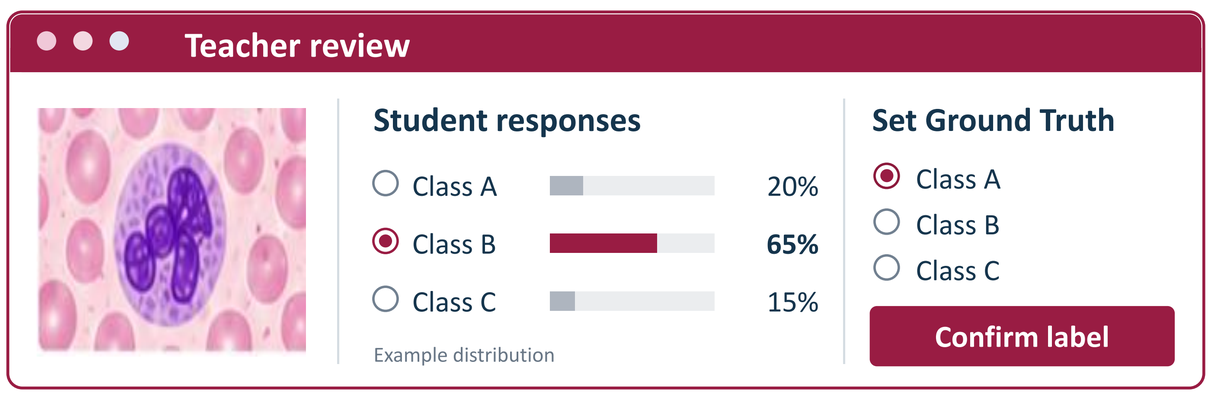
Flexible data collection

Classes can be added or revised by species and teaching objective. Verified labels retain species information and coordinates in the source image.



03 STUDENT & TEACHER VIEWS



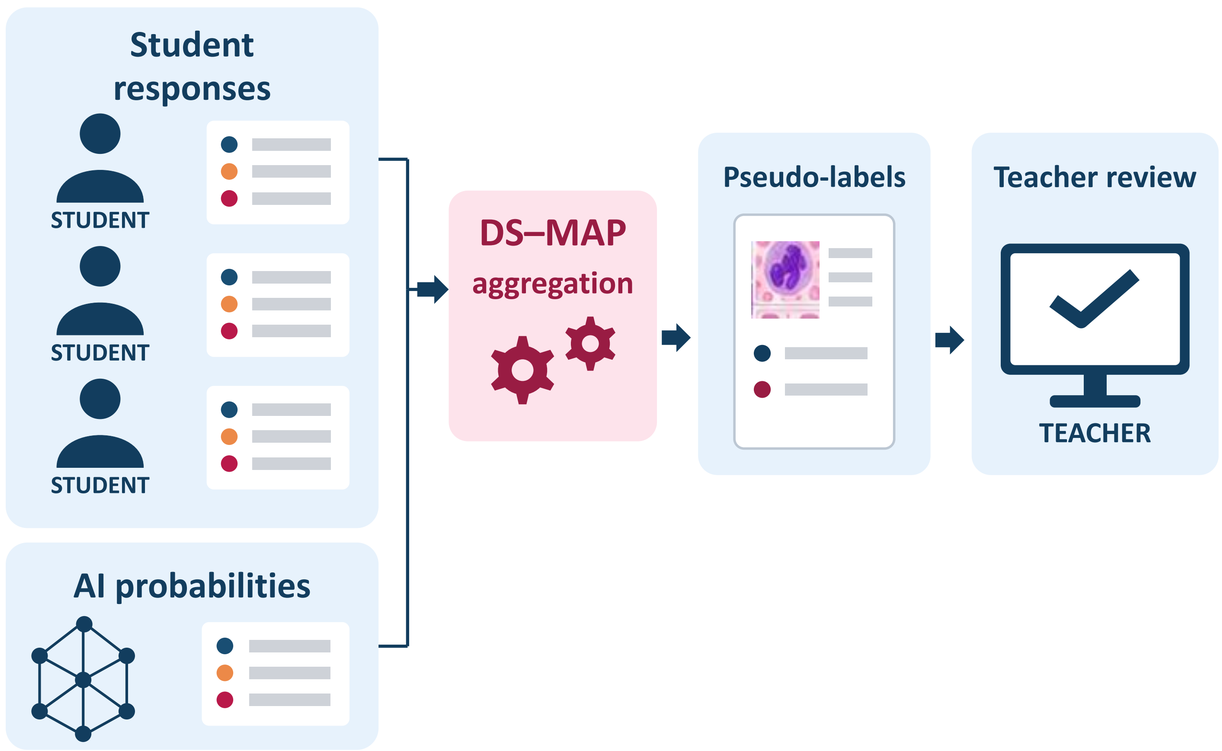


Illustrative interface concepts, not actual screenshots or experimental data.

04 DS–MAP LABEL AGGREGATION

Student reliability with AI prior information

Dawid–Skene (DS) estimates each student's confusion matrix and response reliability. Maximum a posteriori (MAP) inference incorporates AI class probabilities as prior information to infer pseudo-labels.



Priority for teacher review

Cases with low AI confidence or disagreement between AI predictions and student responses.

Teachers confirm the final reference labels.

05 EVALUATION PLAN

Analyze student responses collected in prior research using teacher-confirmed labels as the reference.

Method	Information used
Majority vote	Student vote counts
AI prediction	AI class probabilities
DS–MAP	Student reliability + AI priors

Evaluation criteria

Accuracy, confidence and the proportion of labels eligible for automatic confirmation.

Can DS–MAP improve pseudo-label reliability and help identify cases that require teacher review?

Quantitative results are not reported in the submitted abstract.

06 EXPECTED APPLICATIONS

Education

Assess learner performance and error patterns to guide teacher feedback.

Dataset development

Combine student responses, label aggregation and teacher verification to accumulate reviewed cell labels.

Connecting morphology education with teacher-verified label data

Keywords: Blood cell morphology; Biology education; Human-in-the-loop; Data labeling